Compressor off-design report: summary and diagnostics

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Rotor-Stator  |  run: Max_PR_Compressor_design_rs_no_last_stator

Rotor-Stator  |  PR_des = 1.8855, eta_des = 0.8255  |  correlations: polynomial, Howell factor 2.00

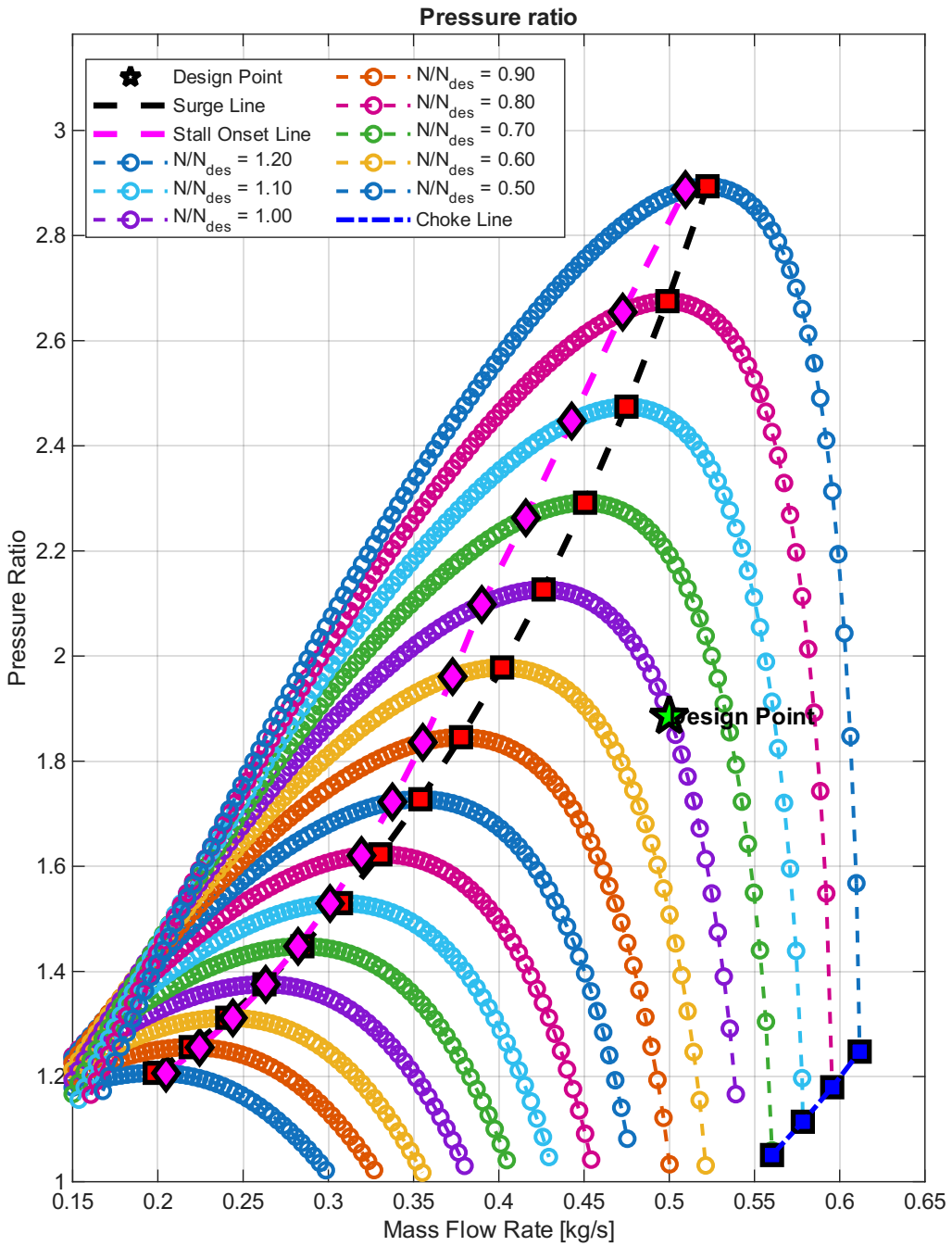
STABILITY (design speed): first limit = peak at m = 0.4263 kg/s (PR 2.127); stall margin SM = 32.3 %
   peak of the speed line (zero slope): m = 0.4263 kg/s | Howell stall onset: m = 0.3900 kg/s
   Koch stage limit: not reached on the computed part of the line
CHOKE (design speed): m_choke = 0.5416 kg/s (Rotor 2 exit (relative)) -> choke margin 8.3 %, operating range (m_choke - m_lim)/m_choke = 21.3 %
SURGE OR ROTATING STALL (design speed): U/(2a_p) = 0.405 -> surge if the installation has V_p/(A_c L_c) > 3.89 (B > 0.80), rotating stall below.
BLADE ROWS at the first limit (peak): highest loss rise Stator 1 (omega/omega_des = 1.58), lowest de Haller Rotor 1 (0.54). Details: Tables > Blade
      rows.

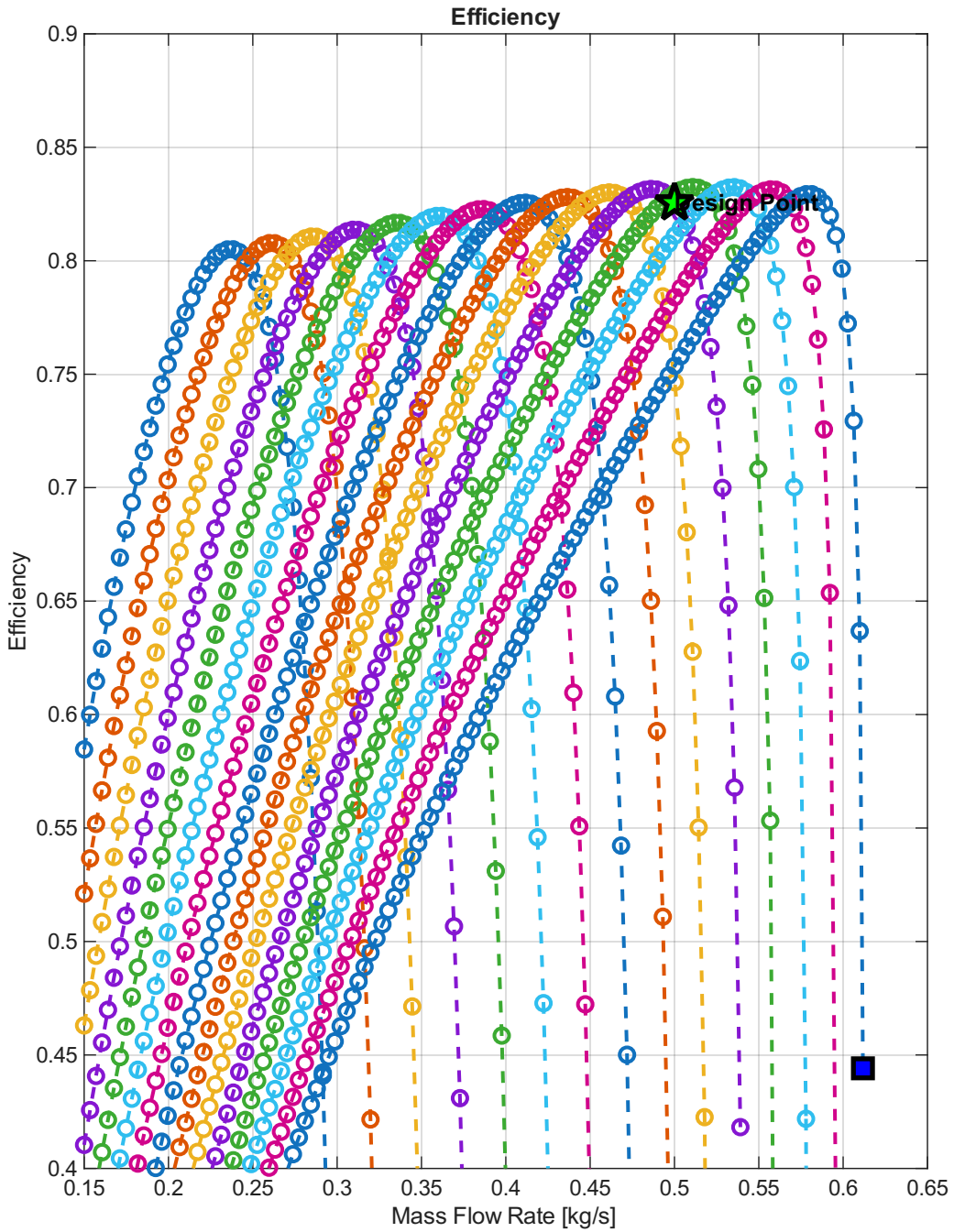
KOCH STAGE-LOADING INDICATOR (never a stability limit): INDICATIVE
   Indicative: near Koch's data. Read the ratio as a trend (more or less loaded than a typical stage), not as a limit. Howell and the peak set the
      limit.
   Koch line: none to draw. No speed line reaches Ch_ef/Ch_max = 1 before the peak or Howell limit, so the stage stays below its Koch limit over the
      whole stable range.
   why: Stage 1 (R1+S1): L/g2 = 2.41 is above 2.2 (high solidity), outside Koch's data, so its stalling pressure rise is extrapolated.
   why: Stage 2 (R2 only): L/g2 = 2.41 is above 2.2 (high solidity), outside Koch's data, so its stalling pressure rise is extrapolated.
Koch check at the design point (target Ch_ef/Ch_max <= 0.85):
  Stage 1 (R1+S1): OK (Ch_ef/Ch_max = 0.73 <= target 0.85)
     - margin available: the solidity could be LOWERED x0.66 (stage L/g2 2.41 -> 1.60): R1 1.97 -> 1.31, S1 1.97 -> 1.31; re-run the design script
      with it (profile loss changes)
     - or RAISE the stage static-pressure rise by up to about 16% before reaching the target
     - note: outside Koch's data: L/g2 = 2.41 (data 0.6-2.2) -> indicative only
  Stage 2 (R2 only): OK (Ch_ef/Ch_max = 0.71 <= target 0.85)
     - margin available: the solidity could be LOWERED x0.63 (stage L/g2 2.41 -> 1.51): R2 1.97 -> 1.24; re-run the design script with it (profile
      loss changes)
     - or RAISE the stage static-pressure rise by up to about 19% before reaching the target
     - note: outside Koch's data: L/g2 = 2.41 (data 0.6-2.2) -> indicative only
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Inputs of this run (.mat data and settings)

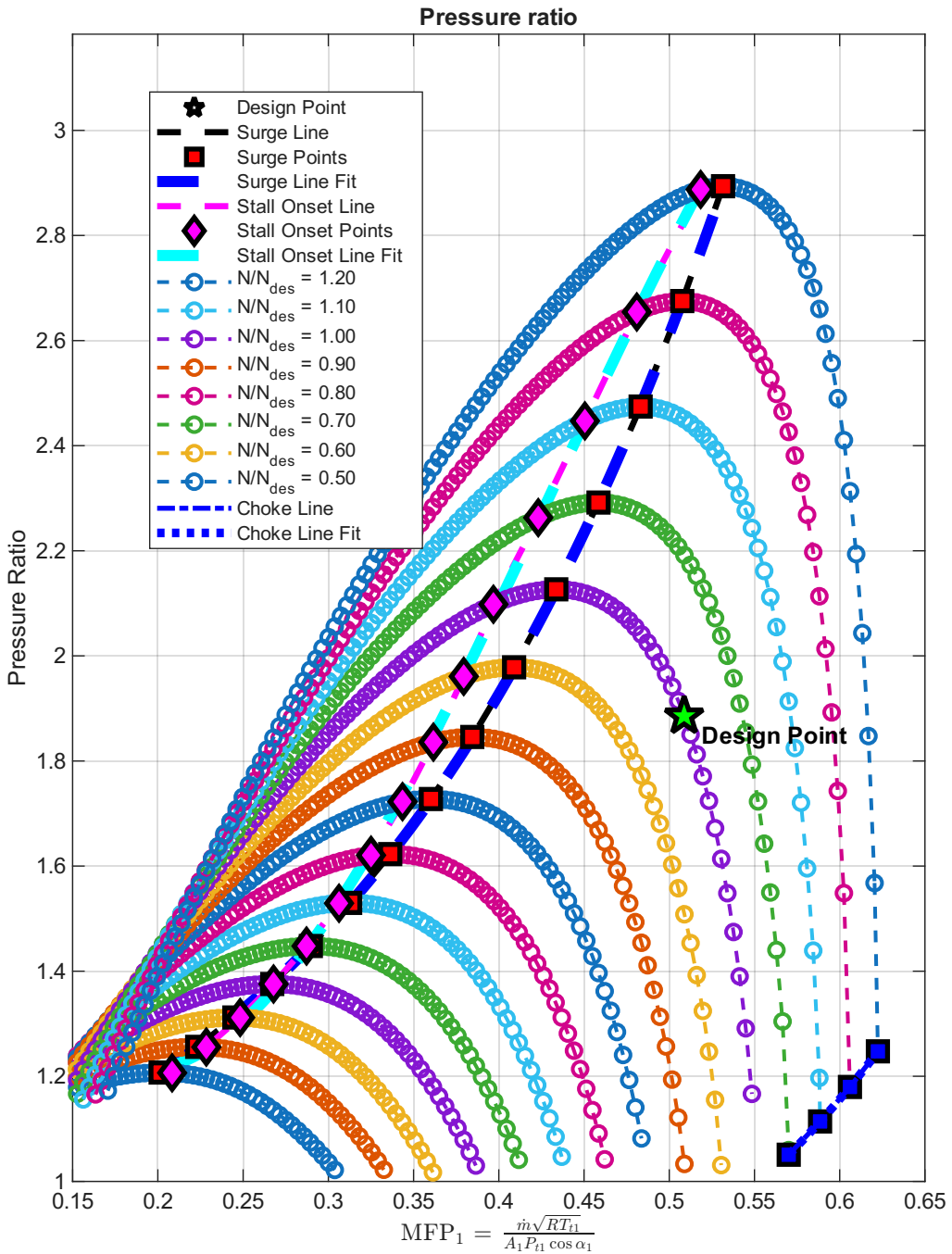
Variable	Value	Unit	Source
Configuration	Rotor-Stator		
Input source	.mat: Max_PR_Compressor_design_rs_n...		
alpha1	38.4033	deg	.mat: Max_PR_Compressor_design_rs_n...
n_stages	2	-	.mat: Max_PR_Compressor_design_rs_n...
A_vec	0.00360564 0.00323586 0.00290087 0....	m^2	.mat: Max_PR_Compressor_design_rs_n...
beta1_dash	51.8472	deg	.mat: Max_PR_Compressor_design_rs_n...
beta2_dash	34.8686	deg	.mat: Max_PR_Compressor_design_rs_n...
m_dot	0.5	kg/s	.mat: Max_PR_Compressor_design_rs_n...
w_loss_rotor	0.140799	-	.mat: Max_PR_Compressor_design_rs_n...
deviation	3.53466	deg	.mat: Max_PR_Compressor_design_rs_n...
incidence	6.87796	deg	.mat: Max_PR_Compressor_design_rs_n...
deflection	20.3219	deg	.mat: Max_PR_Compressor_design_rs_n...
w_loss_stator	0.115656	-	.mat: Max_PR_Compressor_design_rs_n...
U	315.356	m/s	.mat: Max_PR_Compressor_design_rs_n...
sigma_rotor	1.97032	-	.mat: Max_PR_Compressor_design_rs_n...
sigma_stator	1.97032	-	.mat: Max_PR_Compressor_design_rs_n...
chord	0.01	m	.mat: Max_PR_Compressor_design_rs_n...
Correlation set	polynomial		Run settings
Howell factor	2	-	Run settings
Design inlet Tt1	288	K	Run settings
Design inlet pt1	100000	Pa	Run settings
Speed lines N/N_des	0.50 to 1.20 (15 lines)	-	Run settings
Mass-flow points per line	135	-	Run settings
Koch target Ch_ef/Ch_max	0.85	-	Run settings
B critical	0.8	-	Run settings
T_ref / p_ref (corrected map)	288.15 / 101325	K / Pa	Run settings
Koch tip clearance eps/g	0.055	-	Run settings
Koch axial gap dz/s	0.38	-	Run settings

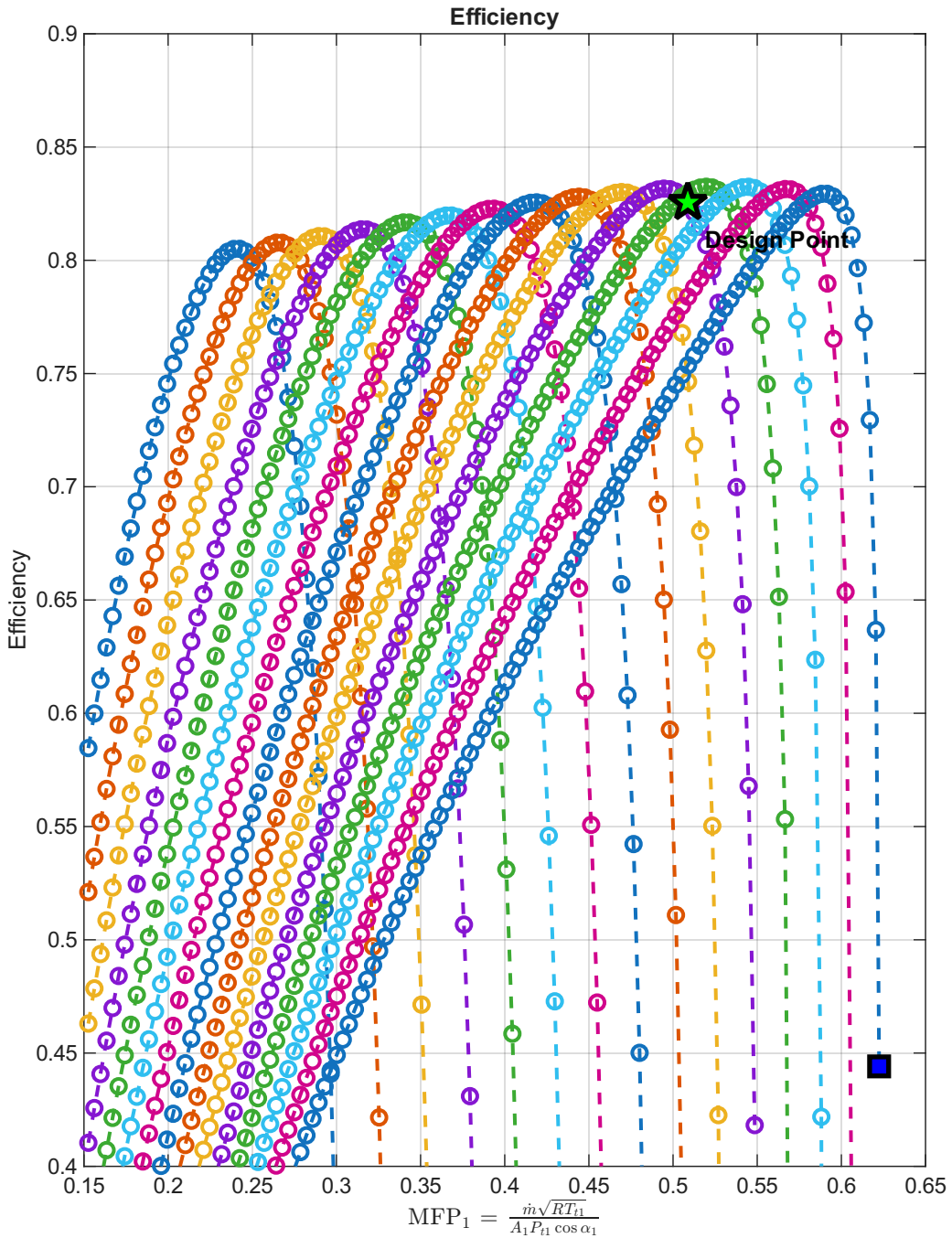
Standard (dimensional) map - Rotor-Stator



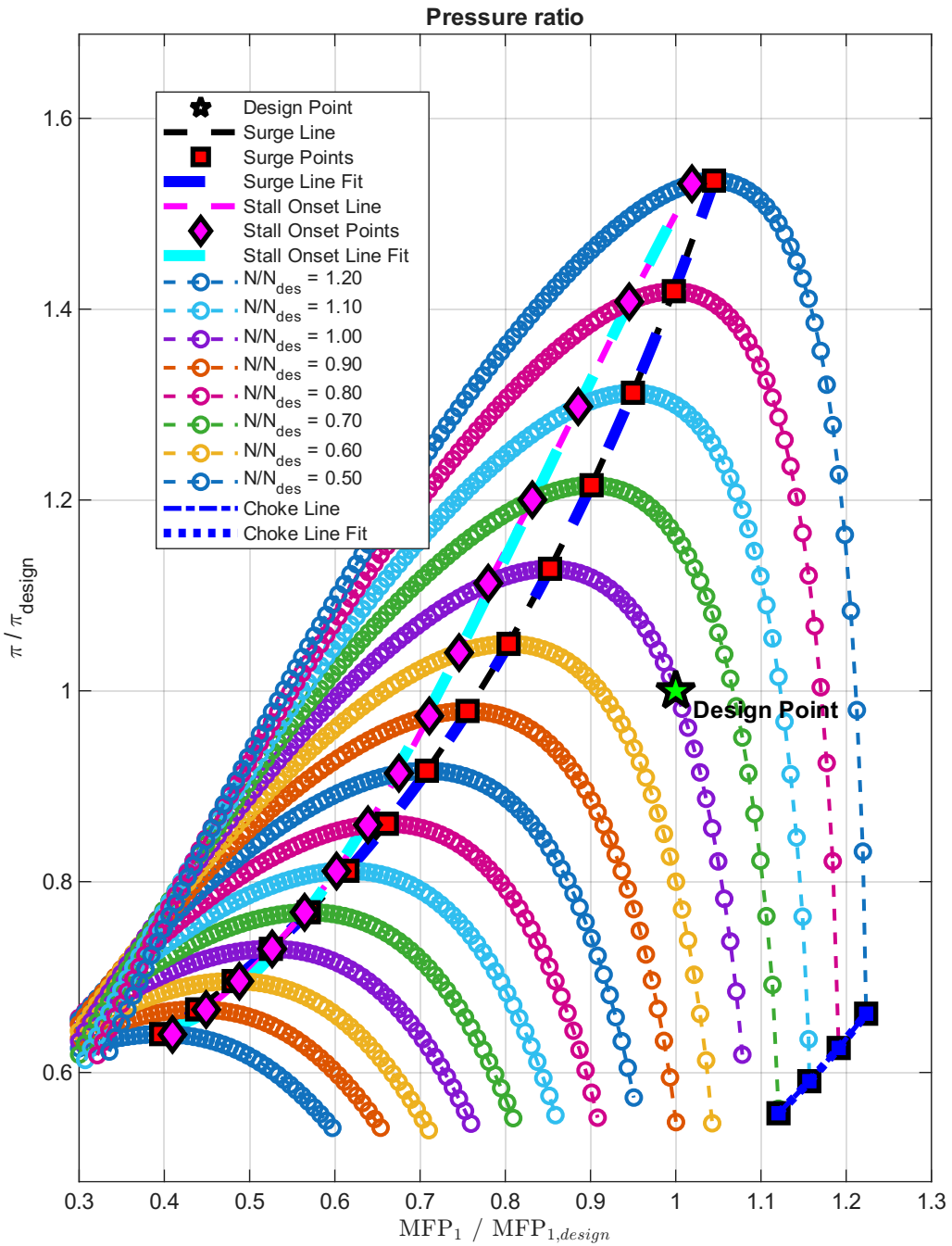


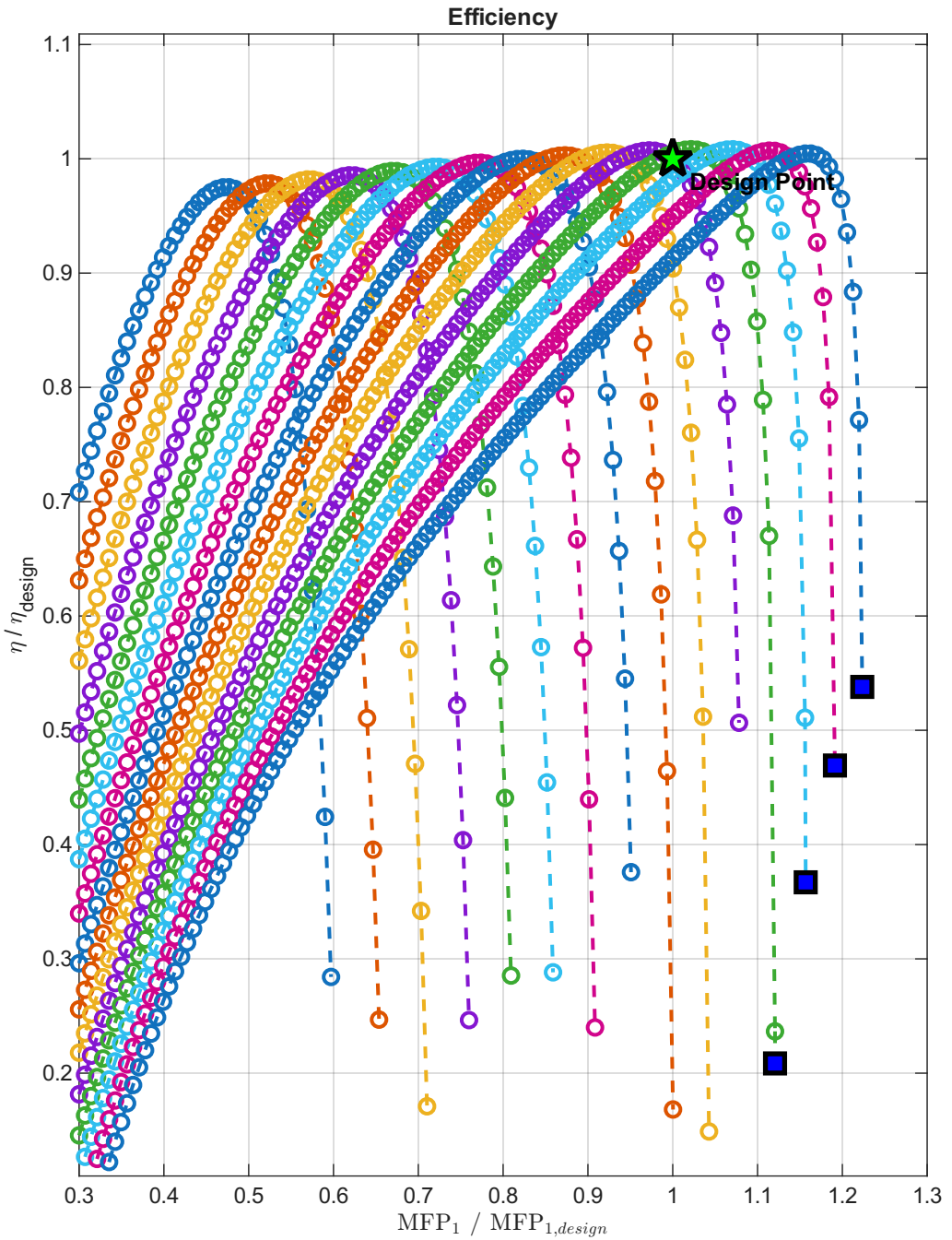
Non-dimensional map - Rotor-Stator



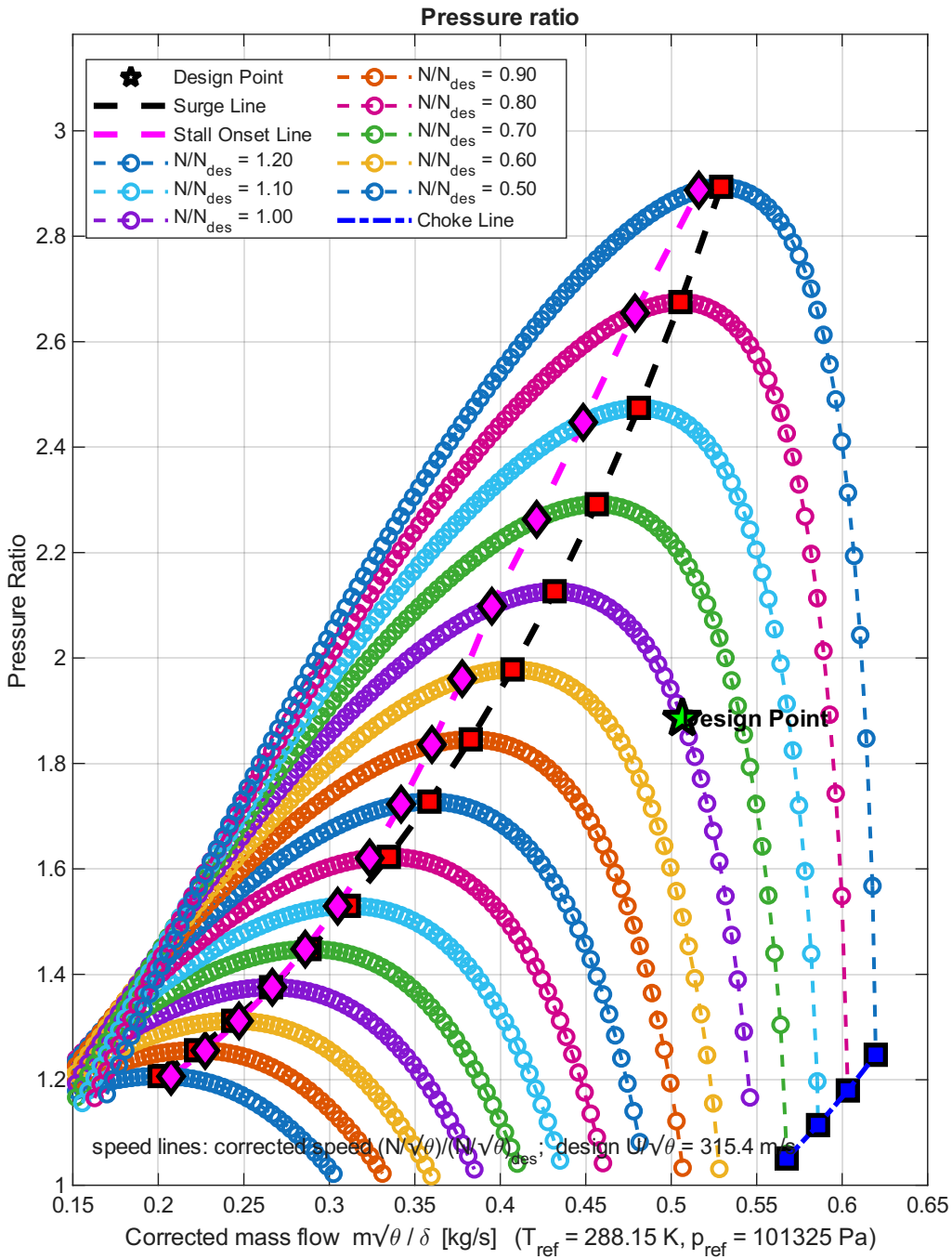


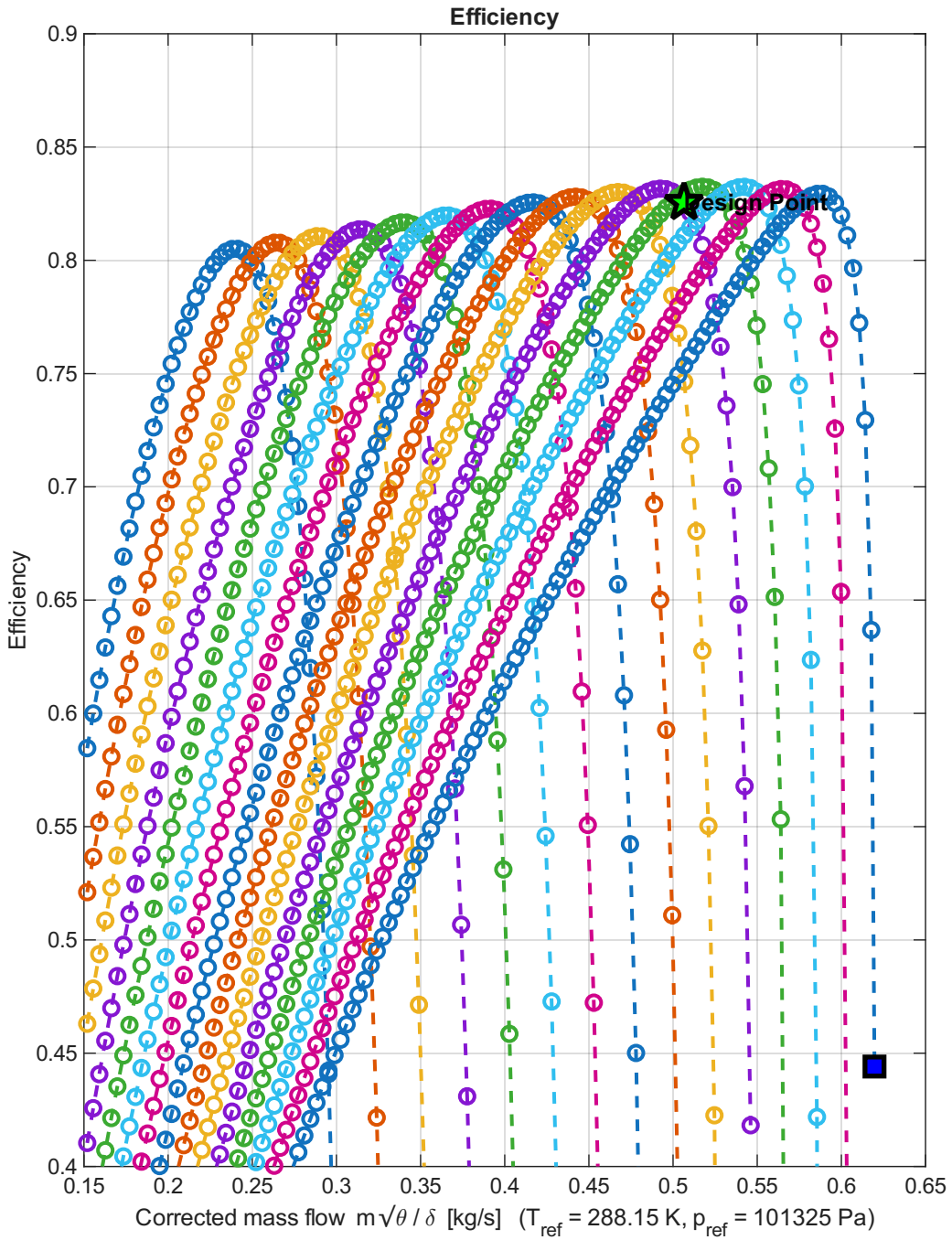
Relative non-dimensional map - Rotor-Stator





Corrected map - Rotor-Stator





Fitted boundary equations (surge, stall onset, choke)

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==== Non-dimensional Compressor Map: Pressure Ratio - Rotor-Stator ====
Surge line:
Surge_PR = 13.4999*MFP^3 + -4.1095*MFP^2 + 2.3130*MFP + 0.7952   [R^2 = 0.99999]
Stall-onset line (Howell):
StallOnset_PR = -38.7824*MFP^3 + 50.4510*MFP^2 + -14.9821*MFP + 2.5026   [R^2 = 0.99963]
Choke line:
Choke_PR = 9.8209*MFP^2 + -7.9557*MFP + 2.3944   [R^2 = 0.99999]
==== Relative Non-dimensional Compressor Map: Pressure Ratio - Rotor-Stator ====
Surge line:
(PR/PR_design) = 0.9428*x^3 + -0.5641*x^2 + 0.6241*x + 0.4217   (x = MFP/MFP_design)   [R^2 = 0.99999]
Stall-onset line (Howell):
(PR/PR_design) = -2.7083*x^3 + 6.9253*x^2 + -4.0424*x + 1.3273   (x = MFP/MFP_design)   [R^2 = 0.99963]
Choke line:
(PR/PR_design) = 1.3481*x^2 + -2.1466*x + 1.2699   (x = MFP/MFP_design)   [R^2 = 0.99999]
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Design point

Quantity	Value
Configuration	Rotor-Stator
Pressure ratio PR_des	1.8855
Isentropic efficiency eta_des	0.82554
Mass flow m_des [kg/s]	0.5
MFP_1,des	0.50874
Corrected mass flow [kg/s]	0.50649
Inlet Tt1 [K] / Pt1 [Pa]	288 / 100000
Correlation set	polynomial
Howell factor	2

Choke

N/N_des	m_choke [kg/s]	MFP_1	PR	Eff	Choking row	Inside compressor region (PR>1)
0.5	0.35979	0.3661	0.582		Rotor 2 exit (relative)	no
0.55	0.37585	0.3824	0.607		Rotor 2 exit (relative)	no
0.6	0.39275	0.3996	0.636		Rotor 2 exit (relative)	no
0.65	0.41031	0.4175	0.668		Rotor 2 exit (relative)	no
0.7	0.42811	0.4356	0.703		Rotor 2 exit (relative)	no
0.75	0.44634	0.4541	0.742		Rotor 2 exit (relative)	no
0.8	0.465	0.4731	0.784		Rotor 2 exit (relative)	no
0.85	0.4841	0.4926	0.829		Rotor 2 exit (relative)	no
0.9	0.50341	0.5122	0.88		Rotor 2 exit (relative)	no
0.95	0.52262	0.5318	0.934		Rotor 2 exit (relative)	no
1	0.54165	0.5511	0.99		Rotor 2 exit (relative)	no
1.05	0.56031	0.5701	1.051	0.172	Rotor 2 exit (relative)	yes
1.1	0.57839	0.5885	1.114	0.303	Rotor 2 exit (relative)	yes
1.15	0.59562	0.606	1.18	0.387	Rotor 2 exit (relative)	yes
1.2	0.61173	0.6224	1.247	0.444	Rotor 2 exit (relative)	yes

Stall onset (Howell)

N/N_des	m_stall [kg/s]	PR	Eff	Stalling row
0.5	0.20478	1.206	0.766	Rotor 1
0.55	0.22449	1.256	0.765	Rotor 1
0.6	0.24399	1.312	0.764	Rotor 1
0.65	0.26326	1.376	0.762	Rotor 1
0.7	0.28227	1.448	0.76	Rotor 1
0.75	0.30102	1.529	0.758	Rotor 1
0.8	0.31947	1.621	0.754	Rotor 1
0.85	0.33762	1.723	0.75	Rotor 1
0.9	0.35544	1.836	0.745	Rotor 1
0.95	0.37292	1.961	0.739	Rotor 1
1	0.39004	2.098	0.733	Rotor 1
1.05	0.41589	2.263	0.738	Stator 1
1.1	0.44279	2.447	0.743	Stator 1
1.15	0.47273	2.654	0.751	Rotor 2
1.2	0.50954	2.888	0.765	Rotor 2

Peak (zero slope)

N/N_des	m_peak [kg/s]	PR	Eff	m_stall - m_peak [kg/s]	First limit on throttling
0.5	0.19815	1.207	0.751	0.00662	stall onset (Howell)
0.55	0.21923	1.256	0.754	0.00526	stall onset (Howell)
0.6	0.24067	1.312	0.757	0.00332	stall onset (Howell)
0.65	0.26251	1.376	0.761	0.00075	stall onset (Howell)
0.7	0.28476	1.448	0.765	-0.00248	peak (zero slope)
0.75	0.30743	1.53	0.768	-0.00642	peak (zero slope)
0.8	0.33052	1.623	0.772	-0.01105	peak (zero slope)
0.85	0.35401	1.728	0.775	-0.01639	peak (zero slope)
0.9	0.37784	1.846	0.778	-0.02241	peak (zero slope)
0.95	0.40196	1.978	0.781	-0.02904	peak (zero slope)
1	0.42627	2.127	0.783	-0.03623	peak (zero slope)
1.05	0.45064	2.292	0.784	-0.03474	peak (zero slope)
1.1	0.47492	2.474	0.784	-0.03212	peak (zero slope)
1.15	0.49892	2.675	0.782	-0.0262	peak (zero slope)
1.2	0.52245	2.894	0.78	-0.01291	peak (zero slope)

Koch limit

N/N_des	m_Koch [kg/s]	PR	Eff	Limiting stage
0.5	NaN	NaN	NaN	not reached
0.55	NaN	NaN	NaN	not reached
0.6	NaN	NaN	NaN	not reached
0.65	NaN	NaN	NaN	not reached
0.7	NaN	NaN	NaN	not reached
0.75	NaN	NaN	NaN	not reached
0.8	NaN	NaN	NaN	not reached
0.85	NaN	NaN	NaN	not reached
0.9	NaN	NaN	NaN	not reached
0.95	NaN	NaN	NaN	not reached
1	NaN	NaN	NaN	not reached
1.05	NaN	NaN	NaN	not reached
1.1	NaN	NaN	NaN	not reached
1.15	NaN	NaN	NaN	not reached
1.2	NaN	NaN	NaN	not reached

Stability limit & Greitzer B

N/N_des	First limit	m_lim [kg/s]	PR	U [m/s]	a_p [m/s]	U/(2a_p)	(V_p/(A_c L_c))_crit	B	Beyond the limit
0.5	stall onset	0.20478	1.206	157.7	352.2	0.2239	12.77	NaN	-
0.55	stall onset	0.22449	1.256	173.4	354.8	0.2444	10.71	NaN	-
0.6	stall onset	0.24399	1.312	189.2	357.7	0.2645	9.15	NaN	-
0.65	stall onset	0.26326	1.376	205	360.8	0.284	7.93	NaN	-
0.7	peak	0.28476	1.448	220.7	364.1	0.3031	6.97	NaN	-
0.75	peak	0.30743	1.53	236.5	367.7	0.3217	6.19	NaN	-
0.8	peak	0.33052	1.623	252.3	371.4	0.3396	5.55	NaN	-
0.85	peak	0.35401	1.728	268.1	375.4	0.357	5.02	NaN	-
0.9	peak	0.37784	1.846	283.8	379.7	0.3737	4.58	NaN	-
0.95	peak	0.40196	1.978	299.6	384.2	0.3899	4.21	NaN	-
1	peak	0.42627	2.127	315.4	389	0.4054	3.89	NaN	-
1.05	peak	0.45064	2.292	331.1	394	0.4203	3.62	NaN	-
1.1	peak	0.47492	2.474	346.9	399.2	0.4345	3.39	NaN	-
1.15	peak	0.49892	2.675	362.7	404.6	0.4481	3.19	NaN	-
1.2	peak	0.52245	2.894	378.4	410.3	0.4612	3.01	NaN	-

Blade rows

Operating point (N = 1)	m [kg/s]	Blade row	Incidence i [deg]	(i - i_des)/deflection	Loss omega	omega/omega_des	M_in (row frame)	M_out (row frame)	de Haller V_out/V_in	Deviation [deg]	Watch
Design point	0.5	Rotor 1	6.88	0	0.1408	1	0.75	0.482	0.662	3.53	de Haller < 0.72
Design point	0.5	Stator 1	6.88	0	0.1157	1	0.727	0.468	0.662	3.53	de Haller < 0.72
Design point	0.5	Rotor 2	6.88	0	0.1408	1	0.707	0.456	0.662	3.53	de Haller < 0.72
First limit (peak)	0.42627	Rotor 1	13.57	0.329	0.2087	1.482	0.76	0.392	0.537	5.75	de Haller < 0.72
First limit (peak)	0.42627	Stator 1	13.55	0.328	0.1831	1.583	0.715	0.375	0.543	5.73	de Haller < 0.72
First limit (peak)	0.42627	Rotor 2	13.87	0.344	0.215	1.527	0.692	0.361	0.539	5.94	de Haller < 0.72
Near choke (0.99 m_choke)	0.53623	Rotor 1	3.04	-0.189	0.1626	1.155	0.749	0.564	0.769	3.47	
Near choke (0.99 m_choke)	0.53623	Stator 1	1.01	-0.289	0.1661	1.436	0.732	0.602	0.835	3.53	
Near choke (0.99 m_choke)	0.53623	Rotor 2	-2.73	-0.473	0.2735	1.942	0.721	0.718	0.996	3.49	